

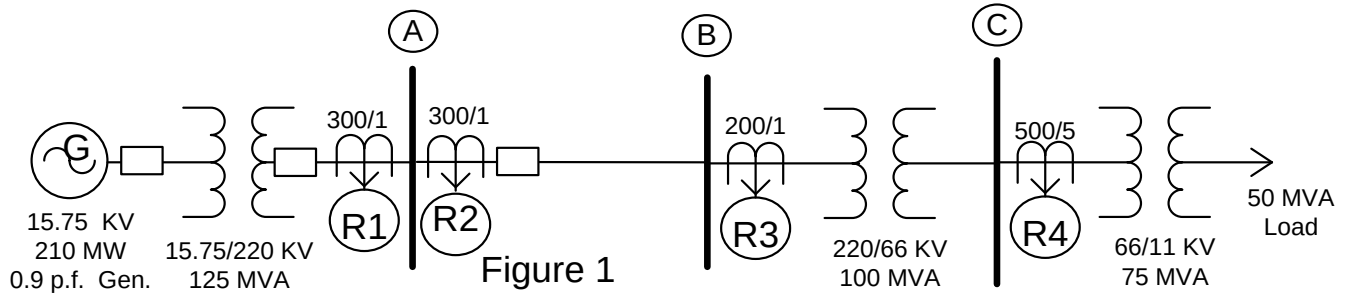
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (EE) Examination****May 2016****EE405 Power System Protection****Date: 11/05/2016, Wednesday****Time: 01.30 p.m. to 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1** Choose the correct option**[07]**

- I. Purpose of backup protection is
 - (a) to increase the speed.
 - (b) to increase a reach.
 - (c) to leave no blind spot.
 - (d) to guard against failure of primary.
- II. The most efficient torque-producing actuating structure for induction-type relays is
 - (a) shaded pole structure.
 - (b) watt-hour meter structure.
 - (c) induction-cup structure.
 - (d) single induction loop structure.
- III. Buchholz relay is
 - (a) located in the conservator tank.
 - (b) located in the transformer tank itself.
 - (c) connected in the pipe connecting main tank of transformer and conservator.
 - (d) installed in the circuit breaker.
- IV. Buchholz relays are used on
 - (a) air cooled transformers,
 - (b) instrument transformers.
 - (c) distribution transformers.
 - (d) oil immersed power transformers of ratings above 500 kVA.
- V. In an overcurrent protection the setting of the earth fault relay is
 - (a) more than the phase fault relay.
 - (b) equal to the phase fault relay.
 - (c) less than the phase fault relay.
 - (d) the two settings are unrelated to each other.
- VI. A fault current of 2,000 A is passing on the primary side of a 400/5 C.T. On the secondary side of the C.T., an inverse-time overcurrent relay is connected whose plug setting is set at 50%. The plug setting multiplier will be:
 - (a) 25
 - (b) 50
 - (c) 10
 - (d) 30
- VII. Directional overcurrent relays have two exciting coils connected across
 - (a) CT secondaries of two different phases.
 - (b) VT secondaries of two different phases.
 - (c) CT and VT secondaries of the same phase.
 - (d) CT and VT secondaries of two different phases.

Q – 2. A Figure 1 shows a single-line diagram of a portion of a power-system network. The relays shown are standard IDMT relays (setting range= 50 to 200% of CT secondary rating). The three phase short circuit current of bus A, B and C are 3000 Amp, 2500 Amp and 6000 Amp respectively. TMS of R_4 is set as 0.1 each. Determine the plug setting of all relays and time setting of R_1, R_2 and R_3 . **[08]**



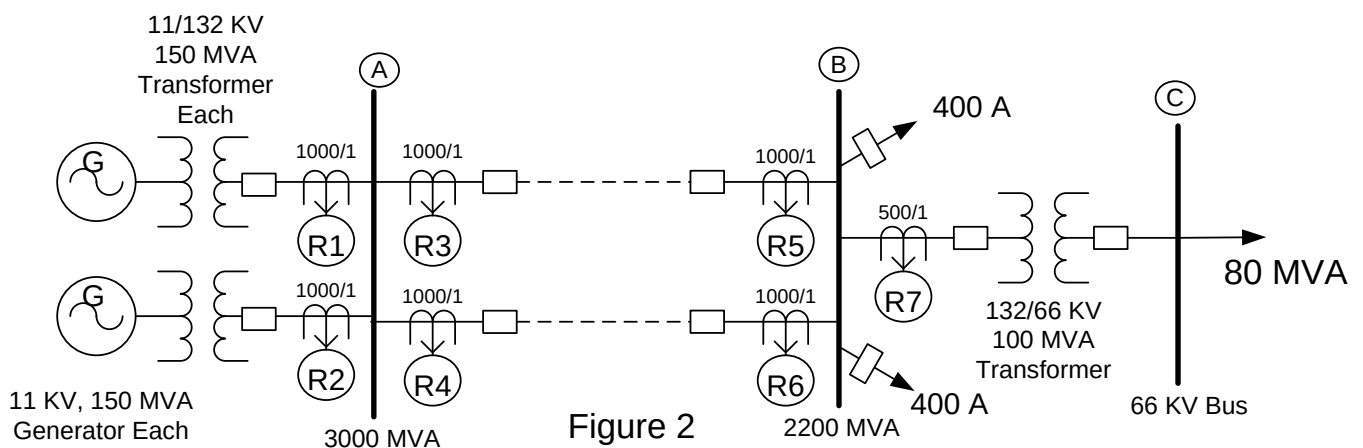
B Give Reason

- i. Two o/c and one E/E protection scheme proves to be inadequate for Transformer connected feeder protection.
- ii. Relays giving very inverse time-current characteristic are more suitable where the ratio of source impedance to line impedance is large.

OR

Q – 2. A Explain with the help of appropriate circuit diagram and vector diagram, 30° , 60° and 90° connection schemes for directional relays. **[06]**

B Figure 2 shows a single-line diagram of a power system. Bracketed values give the fault levels on the concerned bus. Relays used are IDMT over current relays. Relays are rated for 1 A with a setting range of 50-200% of 1 A in 7 equal steps each of 20%. Relay R7 is set at a TMS of 0.25. Assuming a suitable discrimination time, determine the plug-settings and time-settings of all the relays shown in Figure 2. [08]



Q – 3. Attempt any **Two**.

- A** The part of the power system is shown in figure 3. Decide the three zone settings of a mho distance relay R having characteristic angle of 60° . First zone covers 85% of transmission line L_2 , second zone reach is up to bus C and third zone covers 70% of transmission line L_3 . [07]

Required data is as follows:

- (i) 220 KV Lines L_1 and L_2
Impedance: $0.06 + j 0.18$ ohm / km / ph. Line length: 150 Km.
(ii) 66 KV line L_3 :
Impedance: $0.04 + j 0.08$ ohm / Km / ph. Line length: 100 Km

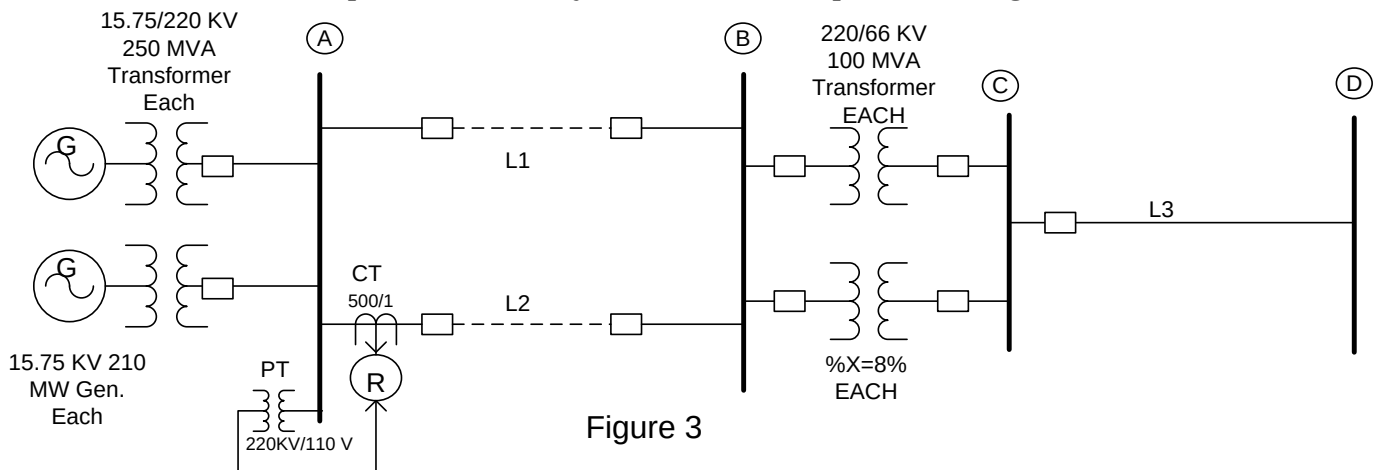


Figure 3

- B** With neat diagram explain the block diagram of carrier current protection scheme. [07]
- C** Derive the expression for estimation of phasors using a full-cycle Fourier algorithm. [07]

SECTION – II

Q – 4. Choose the correct option [07]

- I. For the protection of lines against faults involving variable fault resistance, the preferred relaying scheme is a
 - (a) plain impedance relay, (b) directional overcurrent relay.
 - (c) mho relay. (d) reactance relay.
- II. Mho relay is usually employed for the protection of
 - (a) short lines only. (b) medium lines only.
 - (c) long lines only. (d) any line.
- III. Differential relays are used for protection of equipment against
 - (a) internal faults. (b) overcurrent.
 - (c) reverse current. (d) reverse power.
- IV. Both voltage and current signals are required for
 - (a) a plain overcurrent relay. (b) a differential relay.
 - (c) a directional relay. (d) a biased differential relay.

- V. Zero-sequence current is used for relaying purposes only in the case of
 (a) phase overcurrent relay. (b) phase impedance relay,
 (c) ground overcurrent relay, (d) ground impedance relay.
- VI. As soon as a fault develops in a generator stator, it is essential to suppress field excitation, otherwise
 (a) terminal voltage will drop.
 (b) it may lead to loss of synchronism.
 (c) it will continue to supply power to the stator winding fault
 (d) all of the above.
- VII. For the protection of a large squirrel cage induction motor against single phase normally
 (a) an overcurrent relay is used.
 (b) a differential relay is used.
 (c) a directional relay is used.
 (d) negative sequence current sensitive relay is used.

Q – 5. A A 13.75 KV, 3 phase, 50 Hz, 50 MVA, star connected generator is protected by an earth-fault relay having 10% setting. If the neutral resistance limits the maximum earth-fault current to 50% of full-load value, determine the value of the resistor and percentages of the winding remain unprotected. CT ratio is 2000/1 A. [07]

B With neat figure explain the generator inter-turn fault protection. [07]

OR

Q – 5. A List out the abnormalities of induction motor. Explain causes and consequences of these abnormalities. Suggest relay for each abnormalities. [07]

B A transformer is protected by biased differential relay. The relevant details are as per following. [07]

i. Transformer:

250 MVA, 220/132 KV, 3 phase, 50 Hz, DY₁, Neutral solidly grounded.

%X=14%

$I_{\text{mag Inrush}} = 10X I_{\text{Rated}}$, $I_0 = 2\%$ of I_{Rated} .

On load Tap changer= -5% to +7.5% in step of 1.25% ,

ii. Current transformers:

L.V. Side: 1500/1A Y-connected, H.V. Side: 750/1 A Y-connected.

Interposing CTs connected on L.V. side. CTs are of class 3

iii. Biased Differential Relay: (Relay rating 1 Amp.)

Basic setting: 10-50% of 1 amp in step of 10%

Bias setting: 10-50% in step of 10%

H.S. inst. setting 800 to 1600% of 1 Amp in step of 100%

Find out the ICT ratio and draw the detailed 3-Φ diagram of protective scheme. Also suggest the relay setting.

Q – 6. Attempt any two.

- A** Draw neat and well labeled diagram for Split-Bus protection scheme. In this diagram show operation of relay during any in-zone fault. [07]
- B** I. Give reason: It is necessary to protect a large generator against heavy unbalanced load. [03]
II. With neat figure explain this statement: “An offset equal to $X_d'/2$ is provided in the field failure relay characteristic.” [04]
- C** List out the “type tests” that are performed on relay. Explain atleast three of them. [07]
- D** Explain the factors to be considered while selecting a protective CT. [07]
